Helm Charts Tutorial (Beginner) — create, deploy (OpenShift), and store

This is the **hands-on tutorial** part.

If you want the “what is Helm / structure / workflow” overview, read:

- `dataengineering/cloud/helm-charts-intro.md`

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1) Goal (what we will build)

We will:

- create a chart
- configure it to deploy **nginx**
- render the YAML (“compile”)
- deploy it to an OpenShift project
- learn 3 simple ways to store/share the chart

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2) Requirements

- Helm installed: `helm version`
- OpenShift CLI installed: `oc version`
- Access to a cluster (login credentials)

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3) Create a starter chart

``bash mkdir -p helm-demo && cd helm-demo helm create hello-chart ``

Chart structure reminder:

Helm chart structure (what’s inside a chart folder)

```
mychart/
  Chart.yaml        (name/version)
  values.yaml       (your settings)
  templates/        (YAML templates)
    deployment.yaml
    service.yaml
    route.yaml      (OpenShift)
  charts/           (optional deps)
```

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4) Configure the app to deploy (values)

Edit `hello-chart/values.yaml` and set a real image. Example:

```yaml replicaCount: 1

image: repository: nginx tag: "1.25"

service: type: ClusterIP port: 80 ```

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## 5) Validate the chart

```bash helm lint ./hello-chart ```

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## 6) Render (“compile”) to YAML locally

This is the safest step before you touch the cluster:

```bash helm template hello ./hello-chart ```

If the output YAML looks wrong, fix `values.yaml` or templates and rerun.

Workflow reminder:
```
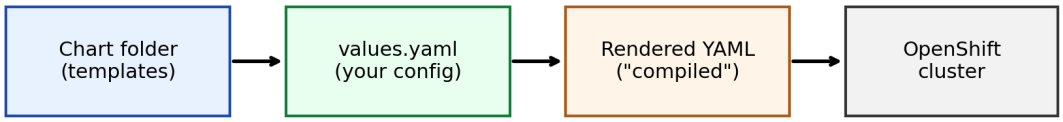

Helm workflow: chart → rendered YAML → cluster



Helm does two jobs: 1) render templates to final YAML 2) apply that YAML to the cluster + store a release record

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7) Deploy to OpenShift

Step A — Login and choose a project

```bash oc login https://api.:6443 oc new-project hello-project

# or

oc project hello-project ```

**Step B — Install the chart into that namespace**

``bash helm install hello ./hello-chart -n hello-project ``

Verify:

``bash helm list -n hello-project oc get all -n hello-project ``

**Step C — Upgrade (after changing values/templates)**

``bash helm upgrade hello ./hello-chart -n hello-project ``

**Step D — Uninstall**

``bash helm uninstall hello -n hello-project ``

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## 8) OpenShift routing: Ingress vs Route (beginner view)

- Kubernetes often uses **Ingress**.
- OpenShift commonly uses **Route**.
```

If you want a URL on OpenShift, you may add a `Route` template.

Example `hello-chart/templates/route.yaml`:

``yaml apiVersion: route.openshift.io/v1 kind: Route metadata: name: {{ include "hello-chart.fullname" . }} spec: to: kind: Service name: {{ include "hello-chart.fullname" . }} port: targetPort: http ``

Tip: teams usually make this optional with a value like `route.enabled: true`.

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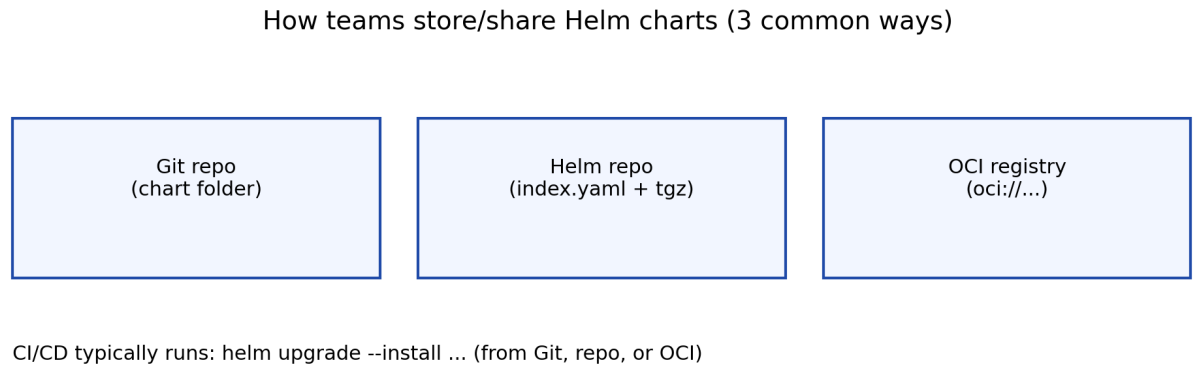
9) How to store/share your chart (3 easy options)



Option 1 — Store chart folder in Git (simplest)

``bash helm upgrade --install hello ./hello-chart -n hello-project -f values-prod.yaml ``

Option 2 — Helm repo (package + index)

``bash helm package ./hello-chart helm repo index . ``

Host the `.tgz` + `index.yaml` (GitHub Pages, S3, Nexus/Artifactory).

Option 3 — OCI registry (modern)

High-level flow:

```bash helm package ./hello-chart helm registry login helm push hello-chart-0.1.0.tgz oci:///charts

helm install hello oci:///charts/hello-chart -n hello-project ```

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```

10) Final mental model

- Chart = blueprint
- Values = settings
- Render (“compile”) = blueprint → final YAML
- Install/upgrade = apply YAML to OpenShift
- Store/share = Git or Helm repo or OCI registry

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11) Your case: JFrog (Artifactory) + OpenShift — every call, step by step

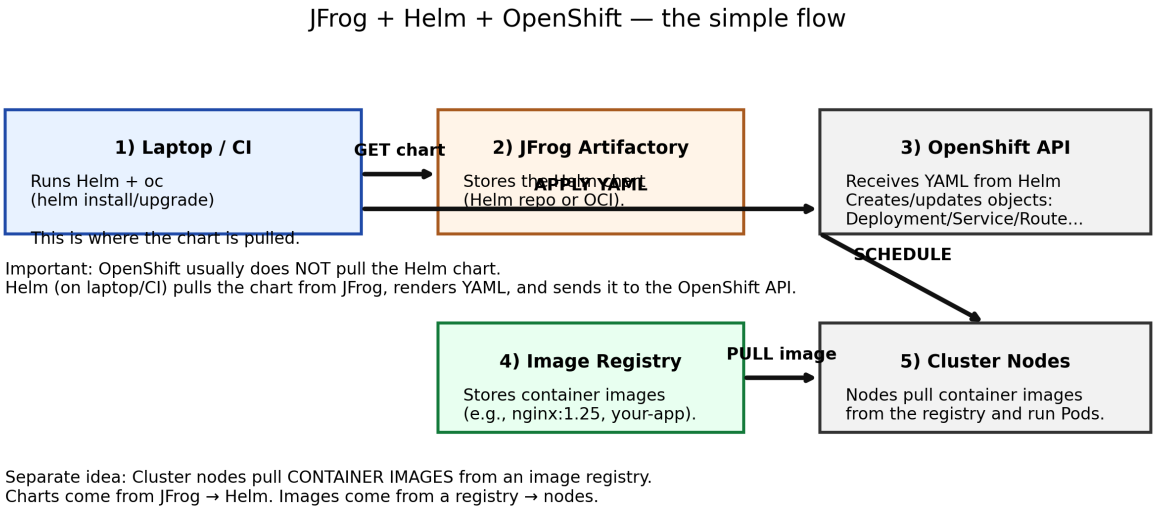
The most common misunderstanding:

The OpenShift cluster usually does NOT pull the Helm chart.

What really happens:

- **Your laptop / CI runner** pulls the chart from **JFrog**.
- Helm renders templates into final YAML (“compile”).
- Helm sends that YAML to the **OpenShift API**.
- The cluster creates Pods, and the **nodes pull container images** from an image registry (not the chart).

Diagram: who talks to who



The calls (in order)

0) Login to OpenShift

``bash oc login https://api.<cluster>:6443 oc project hello-project ``

1) Get the chart from JFrog (two common ways)

A) If JFrog is a Helm repo

``bash helm repo add myjfrog https://<jfrog>/artifactory/api/helm/<helm-repo> helm repo update helm search repo myjfrog ``

Then install by name:

``bash helm upgrade --install hello myjfrog/hello-chart -n hello-project -f values-prod.yaml ``

B) If JFrog is an OCI registry

``bash helm registry login <jfrog> helm pull oci://<jfrog>/<oci-repo>/hello-chart --version 0.1.0 ``

Then install from OCI:

``bash helm upgrade --install hello oci://<jfrog>/<oci-repo>/hello-chart -n hello-project -f values-prod.yaml ``

2) Render (“compile”) happens locally

Helm renders templates locally (CI/laptop):

``bash helm template hello ./hello-chart ``

(If installing from a repo/OCI, Helm does the same rendering step internally before applying.)

3) Apply YAML to the OpenShift API

This happens when you run:

```bash helm install hello ... -n hello-project

# or

helm upgrade --install hello ... -n hello-project ```

**4) Cluster pulls container images**

Then the cluster creates Pods. The nodes pull the **container image** (e.g., `nginx:1.25` or your app image) from your image registry.
```